

Initiation Memo and Task Register

Companion to Section 3. Five-agent panel, cost-ordered funnel, thirty-seven tasks, four gates · 15 September 2026

This memo starts the build. It fixes the rules before the first stage runs, seeds the ledger, names the threats that stop conditions may reference, charters the five agents by evidence access, and lists every task with an owner and an exit criterion. The ordering follows the funnel. Cost rises downward, so the register front-loads the checks that cost nothing and can close the project.

THE ONE RULE THAT GOVERNS THE REST

Four of the six conditions cost no compute. Two gates, in week one and week two, can close this project before a single agent call. Nobody runs tier 3 work until both gates pass. In the protein record every expensive failure sat downstream of an unrun check above it.

PART A. THE CONSTITUTION, PREPENDED TO EVERY STAGE

Rules accumulated across stages and applied inconsistently produced the worst failure in the protein record, where a stop condition in one prompt contradicted a standing rule written two stages earlier and the prompt won. The twelve clauses below prepend automatically to every stage prompt. The Adversary agent checks compliance clause by clause. A prompt that contradicts a clause fails validation before it runs.

1. Count the scarce unit, not the abundant one. Cluster before counting. Report every axis together or report none.
2. A tool must measure the quantity the task scores on. Justify a substitute against that quantity, never against the tool it replaces.
3. The served checkpoint belongs to the tool's identity. Record its hash, parameter count and revision on the card.
4. No tool that helps produce an answer may also score it.
5. Build and measure the comparator before installing the foundation model.
6. Gates fire on the protected quantity, stated as a threat, never on a tool name.
7. A gate arms only after both its must-fire and its must-not-fire control have run.
8. Hash what the runner emits, compare against the specification, and block the run on divergence.
9. Report the decision metric with its chance baseline and with the strongest existing adjudicator. Publish MDE(N) beside every effect.
10. Invalid answers are coverage failures, never zeros. The repair budget stays identical across arms and gets declared in advance.
11. Arms differ only in grant. Verify every grant from the model's side.
12. Preregistered values live in the ledger. Reports read the ledger rather than restating it.

PART B. AMENDMENT LEDGER, SEEDED

Prose preregistration drifts. Every preregistered element below becomes a record with a current value and an amendment chain. Reports read the current value. Drift becomes impossible rather than merely detectable.

Element	Value at initiation	Amendment rule
delta	0.05 precision at twenty	Amendable only before gate 2, and only upward with a stated cost argument
k	20	Frozen. Tied to a real assay batch
n_cand	500	Amendable at item-design search, which runs on the design split only
tau, clustering	Foldseek TM-score 0.50 primary, CD-HIT 40 percent secondary	Frozen after T1.3. Sensitivity reported at 30 and 50 percent identity
Metric primacy	precision at twenty primary, NDCG secondary	Frozen before the run. The protein case saw the two disagree in sign
C, channel count	to be measured at T2.4	Set by mechanical ablation, never asserted
S, samples per cell	5 provisional	Set by the coverage curve at T4.3, then frozen
Repair budget	to be set at T4.3	Fixed, not adaptive, and identical across arms
Label evidence standard	functional evidence only	Frozen. Computational evidence codes do not qualify
sigma_d, N_min, MDE	to be measured at T2.5	Recomputed at the realised N and republished with every effect

PART C. THREAT TAXONOMY FOR STOP CONDITIONS

In the protein record a stop condition written as a tool name blocked a run over two inert search tools and contradicted a standing rule. Every stop condition in this build reads as a predicate over one of the eight named threats below. A name-based condition fails validation at authoring time.

Threat	Definition	Why it matters here
T1	Execution outside the cell sandbox	Standard platform protection. Do not duplicate it in a gate

T2	Network reach beyond the allowlist	An arm fetching a PHI-base record live bypasses the frozen cohort
T3	Package install at run time	Changes the tool surface mid-run and invalidates the card
T4	Filesystem write outside cell scratch	Cross-cell contamination through shared state
T5	Grant violation, meaning an arm reaching a capability its condition withholds	Arm 3 touching a foundation model endpoint destroys the only comparison the benchmark exists to make
T6	Label leakage, meaning any read of an item's own evidence record or of the answer key	The dominant domain threat. Positives carry their own PHI-base entries, which state the assay
T7	Cross-cell memory persisting between cells of one run	Persistent agent memory is a contamination channel. Disable it and verify from the model's side
T8	Scorer contact by any arm	A tool that helps produce an answer must not score it

PART D. MUST-FIRE CONTROL SEEDS

Generated controls go trivial unless at least one control per gate derives from a real prior failure. These five seeds come from the protein record and from this domain's predicted traps. Each gate inherits at least one.

Seed	Real failure it encodes	Control the gate must pass
S1	A sequence-model endpoint served an 8M checkpoint while the floor used 650M scores, with rank correlation 0.558 and sign flips	Must fire when the served parameter count disagrees with the card
S2	A frozen prompt manifest measured 973 bytes while the runner sent 2,023, and two harnesses were compared on different tasks for a full stage	Must fire on any hash divergence between specification and emission
S3	A device-utilisation ceiling aborted the success signal it was meant to protect	Must not fire on a legitimate high-utilisation cell
S4	A heading capitalisation check discarded three valid cells	Must not fire on case or whitespace variation within a valid submission
S5	Domain prediction. A pLDDT threshold discards small disulfide-rich effectors, which is the exact profile of a true positive	Must not fire on a low-confidence known effector. pLDDT stays a covariate

PART E. PANEL CHARTER. WHAT EACH AGENT HOLDS, AND WHAT EACH IS DENIED

The asymmetry is the mechanism. In the protein record every factual error came from the agent holding system state, and none came from the critic sharing the proposer's context. Denials below matter as much as holdings.

Agent	Holds	Denied	Proposed human owner
1. Curator	PHI-base, UniProt, MycoCosm, RNA-seq archives, CD-HIT, Foldseek, ESMFold	the scoring code, every arm output	Xin Dai, with Qun Liu on label eligibility
2. Instrument	the filesystem, served endpoints, tool source, run traces	the label file	Gilchan Park with Joseph Kilgore on the agent system
3. Floor	the fixed tools and the held-out design split	the scored cohort, every agentic arm output	Shinjae Yoo with Xin Dai
4. Referee	the preregistration, the ledger, the scorer	arm identity, tool traces	David Park
5. Adversary	traces, gate declarations, literature, served configuration	authoring items or floors of its own	Graham Neubig and the CMU side

Human owners are proposed from the project record and need confirmation at the first weekly.

Convening rule for the verifier panel

The panel convenes at each gate and aggregates disagreement rather than opinion. A finding raised only by an agent sharing the proposer's context gets logged as a reasoning finding and goes to discussion. A finding raised by an agent holding system state, meaning the Instrument or the Adversary, gets treated as a factual finding and blocks the gate until resolved. Two verifiers that read the same source count as one.

Decision rights

- The Floor agent produces the number that closes the project. The Referee rules on it. Neither may see an agentic arm before ruling.
- The Instrument agent blocks any run on a hash divergence. That block is not a review finding and does not go to discussion.
- The Adversary agent refuses to arm a gate whose two controls have not both run.
- Only the Referee amends the ledger, and every amendment carries a reason and a date.

PART F. TASK REGISTER

Week 0. Setup. Zero compute

ID	Task	Owner	Card	Exit criterion
T0.1	Ratify the twelve-clause constitution and wire it into the stage template	Referee	E1	Every stage prompt carries the clauses automatically. A contradicting prompt fails validation
T0.2	Open the amendment ledger with the ten seeded elements	Referee	E2	Ledger under version control. Every element has a current value and an empty chain
T0.3	Publish the eight-threat taxonomy and rewrite every stop condition as a predicate	Adversary	B2	Zero name-based stop conditions survive validation
T0.4	Stand up execution-time provenance hashing in the runner	Instrument	A3	A deliberately mutated prompt blocks the run automatically
T0.5	Fetch and checksum every asset: PEACE repo and checkpoint, Predector container, PHI-base, MycoCosm bulk	Curator	-	Every asset present with a recorded checksum and licence status

Week 1. Tier 1. Free filters. No agent calls

ID	Task	Owner	Card	Exit criterion
T1.1	Axis enumeration in one pass across all five axes	Curator	D2	One table, five counts, and a refusal to release any single count alone
T1.2	Build the union positive set from PHI-base, PEACE curated CSVs and Predector's confirmed set	Curator	-	Deduplicated FASTA with per-sequence provenance and evidence code
T1.3	Structural clustering. Predicted models for the union set, Foldseek all versus all, cluster at TM 0.50	Curator	D2	Independent unit count at TM 0.50 reported together with sequence counts at 30, 40 and 50 percent
T1.4	Label eligibility filter. Retain functional evidence only	Curator, Adversary reviews	-	Every retained positive names an assay. Computational codes removed and counted
T1.5	Contamination audit. Publication and deposition date per positive against every subject cutoff	Curator	D1	Pre-cutoff and post-cutoff strata sized and dated
T1.6	Tool coverage count. In planta RNA-seq, repeat annotation, assembly quality per species	Curator	D2	Per-channel coverage fraction across the candidate cohort
GATE 1	Independent units at or above 45 after structural clustering, with functional labels, and at least four channels covering most of the cohort. Otherwise restate delta at 0.10 and say so, or close the task.			

Week 2. Tier 2. The floors and the decision

ID	Task	Owner	Card	Exit criterion
T2.1	Build and freeze the design split that never gets scored	Floor	D3	Split hashed and recorded before any floor runs
T2.2	Run Floor C, Predector, blinded, on design-split panels	Floor	-	Ranked output per panel, arm identity withheld from the Referee
T2.3	Run Floor M, PEACE, on the same panels with the checkpoint verified	Floor, Instrument verifies	A3	Ranked output plus a served parameter count matching the card
T2.4	Build Floor M plus six channels and measure C by mechanical ablation	Floor	-	The channel count at which the fixed ensemble stops improving
T2.5	Compute sigma_d, N_min and MDE(N) for both cohort geometries	Floor	-	All three written to the ledger before any agent call exists
T2.6	The headroom test. Global channel weights against per-lifestyle oracle weights	Floor	-	H reported with an interval
T2.7	Item-design search over budget, channel set and contract shape, optimised for floor separation	Floor	D3	Chosen design plus a search log showing the scored cohort was never touched
GATE 2	H exceeds delta, and the six-channel floor does not take the ceiling. Otherwise close the task with zero agent calls spent. This is the gate that decides the project.			

Weeks 3 to 5. Tier 3. The instrument

ID	Task	Owner	Card	Exit criterion
T3.1	Generate roughly eighteen tool cards. Read the code first, then the defining paper	Instrument	A2	Every card carries a disagreement list, explicitly empty or non-empty. Silent reconciliation fails review
T3.2	Clear licences and rule on substitutions for SignalP, TMHMM and Phobius	Instrument	-	Each substitution justified against precision at twenty, with the recall spread quoted

T3 . 3	Construct validity check before functional certification for every tool	Instrument, Adversary reviews	-	Each tool states the quantity it returns and its relation to the scored quantity
T3 . 4	Curate the effector fold library for the structural channel	Curator with Instrument	-	Named folds, source structures, and a written statement that pLDDT is a covariate
T3 . 5	Containerise four arms in Harbor on Predator's existing Nextflow and conda packaging	Adversary, CMU	-	One command runs a cell end to end on either site
T3 . 6	Generate must-fire and must-not-fire controls for every gate, inheriting seeds S1 to S5	Adversary	B1	No gate arms until both of its controls have run
T3 . 7	Verify grants from the model's side for arms 1 and 3, including memory disablement	Adversary	B2	Arm 3 attempts a foundation model call and receives a recorded firewall refusal. T7 verified from the model's side

Week 5. Tier 4. Harness control, run before anything else in this tier

ID	Task	Owner	Card	Exit criterion
T4 . 1	Harness control. One panel, one known-good model, full tool surface	Adversary	-	Contract-compliant submission with per-candidate evidence. Failure here indicts the surface, never the subject
T4 . 2	Rule on prompt symmetry against prompt neutrality and record it	Referee	E2	A written ruling naming the fabrication risk that file paths create for arm 1
T4 . 3	Measure tokens, wall time and coverage per cell. Set S and the repair budget from the curve	Instrument, Referee sets values	C1	S and the repair budget written to the ledger, sized against the actual token allocation
GATE 3 The harness control passes and the cost per cell fits the allocation at the chosen S. A failure here loops back to tier 3, never to the subject model.				

Weeks 6 to 8. The run

ID	Task	Owner	Card	Exit criterion
T5 . 1	Freeze the preregistration and close the ledger for the run	Referee	E2	Hash recorded. Later amendments append rather than overwrite
T5 . 2	Build the scored panels with disjoint structural clusters and same-organism decoys	Curator	-	Panel manifest with cluster IDs and an automated check that no cluster repeats
T5 . 3	Build saprotroph-decoy control panels to estimate the unknown-positive background rate	Curator	-	A background contamination estimate that qualifies every absolute precision figure
T5 . 4	Run four arms at the frozen S with an identical repair budget	Instrument runs, Referee scores blind	C1	Coverage reported as a curve in S per arm. No arm scored zero for a wiring reason
T5 . 5	Score process metrics beside outcome, against a forced-depth control at matched compute	Referee	C2	Evaluation depth, explicit rejection and disagreement handling, each with its own denominator
T5 . 6	Report precision at twenty against chance, against Floor M plus, and with MDE at the realised N	Referee	-	No effect appears without its MDE. A difference below MDE gets reported as no result in either direction
GATE 4 Every arm measured or explicitly declared unmeasured with a reason. In the protein record an entire arm went unscored because a precision-only extraction path was never wired in, and this gate exists for that.				

Standing tracks, running in parallel from week 1

ID	Task	Owner	Card	Exit criterion
TL . 1	Live cohort agent. Scheduled ingest of newly published validated effectors under frozen eligibility and clustering rules	Curator	D1	First refresh produces a dated vintage. Results never pool across vintages without a statement
TV . 1	Convene the asymmetric verifier panel at every gate	All five	A1	Disagreement aggregated, not opinion. Verifiers sharing a source counted once
TV . 2	Maintain the failure ledger. Every failure classified into one of the five classes with its governing rule	Adversary	E1	Each new failure either matches a class or forces a constitution amendment

PART G. CAPACITY AND COST

Item	Estimate	Basis and caution
Predicted structures for the union positive set	under 8 GPU hours	Roughly 900 short sequences. Cheap and it gates the whole project
Foldseek all versus all	minutes	Trivial at this scale
Predector across the scored cohort	about 13 hours, one 16-core node	60,000 proteins at the 4,514 per hour rate the paper reports
PEACE embeddings across the cohort	1 to 3 GPU days	ProtT5-XL with eight dropout views per sequence. Batch size dominates
Agent tokens, arms 3 and 4	measure at T4.3	Do not assume. Cells reading several channels over a 500-protein panel can run an order of

BUDGET LEVER, STATED IN ADVANCE

The cohort can halve and still resolve delta. MDE at 120 panels sits near 0.029, at 60 panels near 0.042 and at 45 panels near 0.048, all below a delta of 0.05. So if the token allocation binds, the correct response cuts panels to 60 and states the MDE, rather than cutting samples per cell and losing the coverage curve. Coverage failure is the one quantity repeated sampling actually fixes, and an arm that fails contract validity at one sample often succeeds at five.

PART H. THE FIRST FIVE ACTIONS

1. **Confirm the panel.** Take the proposed human owners in Part E to the weekly and settle them. The build stalls if the Instrument and Adversary roles sit with the same person, because that collapses the asymmetry the panel exists to create.
2. **Start T1.3 immediately.** Structural clustering of the union positive set gates everything downstream and costs under a day of GPU time. Until that number exists, no supply claim in this domain means anything.
3. **Ask Xin Dai for the PEACE checkpoint and curated CSVs, and record the served parameter count.** The protein record lost a full stage to a checkpoint eighty-one times smaller than the one the floor used.
4. **Put PEACE against Predector on a common partition.** One afternoon of CPU. It produces the floor the benchmark needs and a result worth publishing on its own.
5. **Send licence clearance for SignalP, TMHMM and Phobius into the queue now.** Licence blocks arrive slowly and they forced the worst instrument substitution in the protein record.

WHAT CLOSES THIS PROJECT, RESTATED

Headroom below delta at gate 2. Fewer than 45 independent panels after structural clustering. A measured channel count of two or fewer. Labels that cannot be sourced independently of the EffectorP curation lineage. Any one of these closes the task, and three of the four become known within two weeks at a cost of roughly one GPU day and one node afternoon.

Cards referenced follow the framework's Section 1.6.1. A1 asymmetric verifier panel. A2 generated tool cards. A3 execution-time provenance. B1 auto-generated must-fire controls. B2 threat-stated stop conditions. C1 coverage at S with a declared repair budget. C2 process reward beside outcome. D1 live cohort refresh. D2 automated axis enumeration. D3 item-design search. E1 executable constitution. E2 amendment ledger.